

Nguyen Vu Thanh Tinh

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SUMMARY

Third-year Computer Science undergraduate pursuing Software Engineering with hands-on experience in building scalable, real-time systems and architecting Agentic AI workflows. Combines technical execution with strong product thinking-deeply curious about user behavior and dedicated to leveraging AI-assisted coding tools to accelerate prototype development.

EDUCATION

International University – Vietnam National University HCMC

Expected Graduation: Sep 2027

Bachelor of Science in Computer Science

- **Achievements:** Second Runner-Up - GDGoC Hackathon VietNam 2026, Academic Encouragement Scholarship – Spring 2025.

TECHNICAL SKILLS & LANGUAGES

Languages & Databases: JavaScript, Python, Java, C, HTML/CSS | MySQL, MongoDB

Frameworks & Tools: Node.js, Express.js, React.js, FastAPI, Tailwind CSS | Git, GitHub, Postman, Antigravity IDE, GitHub Copilot

Core Concepts & English: Machine Learning Fundamentals, Computer Vision, AI Prompting | English: Intermediate

PROJECTS

SalinAI – Agentic AI IoT System for Smart Agriculture | *Team of 3*

Apr - May 2026

React.js · Node.js · Express.js · MongoDB · Firebase · LangChain · OpenAI (RAG) · Socket.io

- **Autonomous Multi-Agent Orchestration:** Architected a 4-agent pipeline (Researcher, Orchestrator, Planner, Evaluator) using LangChain.js to automate hardware control decisions with a reasoning latency of 3–7 seconds.
- **RAG-Powered Intelligence:** Engineered a RAG pipeline using MongoDB Atlas Vector Search to synthesize irrigation guidance from technical documents, replacing static rules with dynamic, evidence-based reasoning.
- **Self-Evolving Policy via RLHF:** Implemented a human-in-the-loop feedback loop that extracts "Lessons Learned" from farmer input to refine decision logic via Policy Memory, enabling autonomous system improvement.

Taskora – Modern Collaborative Kanban System | *Personal Project*

Mar 2026 - Present

React.js · Node.js · Express.js · MongoDB · Socket.io

- **Seamless Team Sync:** Engineered a high-concurrency engine using Socket.io Rooms to eliminate communication delays, ensuring instant UI updates for all board members without manual refreshing.
- **Optimized Performance:** Reduced API response time for complex card migrations by implementing concurrent database operations.
- **Security Architecture:** Architected a robust security layer featuring JWT-based authentication and custom RBAC middleware to manage granular permissions for workspaces.

Silkworm Anomaly Detection System | *Academic Research (Ongoing)*

Feb 2026 - Present

Python, Machine Learning, Computer Vision, U-Net, YOLO, SAM

- **Autonomous R&D & Literature Review:** R&D & Literature Review: Investigated academic literature to identify research gaps and independently analyzed anomaly detection workflows.
- **Data Preprocessing:** Independently managed image preprocessing and data augmentation to ensure dataset readiness for testing.
- **Faculty Alignment & Reporting:** Directly communicated research progress, documented technical findings, and presented milestone reports to a university faculty advisor.

ACTIVITIES & INTERESTS

Volunteering: Dedicated participant in university volunteer campaigns focused on community welfare.

Interests: Football, Volleyball, Badminton, and staying current with technical literature.